EPSR-test-ex41 readxyz: cell from the file, many elements, error paths

Program(s): readxyz | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `readxyz` [1] when the second line of the `.xyz` file defines the cell (`<word> a b c alpha beta gamma`), all branches of the default-parameter selection (H, O, N, C and other elements), the end of the coordinate list at a short line, the fallback to stdin when line 2 is not numeric, and the two error exits that write no `.ato` file.

Code exercised

Fortran (original, fortran_src/):

- `readxyz.f`: `run_readxyz` (command line, file search), `readxyz` (parsing of the `.xyz` file, cell from line 2 or stdin, one molecule per atom)
- `auxiliary_routines.f`: `convertalphabetagamma` (a b c alpha beta gamma -> EPSR cell angles)
- `ato_manipulation_rev4.f`: `ctype`, `ftype` (atom types with default parameters), `centreofmass`
- `ato_read_and_save.f`: `moloutc` (writing the `.ato`, non-cubic header)

Rust (port, rust/epsr-core/src/):

- `tools/readxyz.rs`: `run`, `readxyz`, `set_type_parameters` (the `ctype` calls)
- `tools/groupc_common.rs`: `convertalphabetagamma`, `setup_lattice`, `centreofmass`, `moloutc`, `chsymbol`/`chatwt` (103 elements)
- `ato.rs`: `State::ctype`, `State::ftype`

Physical system

Hand-made coordinate sets, no physical simulation. `mixture.xyz`: 12 atoms of 9 elements (N, H, Na, Cl, C, O, K, Br, F; an NH3-like group, a CO2-like triple and isolated atoms) in a triclinic cell 18.5 × 19.0 × 21.2 Å, 88°, 92.5°, 105°. `titled.xyz`: a water molecule and an S atom (4 atoms) in a hexagonal cell 12 × 13 × 14 Å, 90°, 90°, 120°. Default parameters: O eps 0.65 kJ/mol, sigma 3.1 Å; N 0.7, 3.2; C 0.8, 3.7; other elements 0.8, 3.2 with the tabulated atomic mass; H 0, 0, mass 2.

Inputs

`readxyz mixture.xyz` (cell on line 2; coordinate list ended by a blank line); `readxyz titled.xyz < titled.stdin` (line 2 has seven non-numeric words -> read error -> cell from stdin); `readxyz badlabel.xyz` (label `Xx` is not a chemical symbol -> 'Problems with .xyz file'); `readxyz short.xyz` (5 atoms announced, 2 given).

How the accuracy was verified

`commands.txt` (4 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. `readxyz`/`readpdb` crash (SIGSEGV) as compiled from the original source, so the reference build adds an explicit INTERFACE block for `ctype` (`rust/tools/build_patched_readxyz_readpdb.sh`); the port behaves like the patched program. Result of the final regression run of the complete port: every file identical in 1 pass.
Outputs compared: `mixture.ato` (12 molecules, EPSR cell angles 105°, 2.866°, 150.735° in the header) and `titled.ato` (120°, 0°, 0°); for the two error cases both programs write no file. Fortran reference: the original `readxyz`/`readpdb` binaries crash (SIGSEGV) at the first call of `ctype`, which has optional arguments but is called with 11 of its 13 arguments without an explicit interface (gfortran passes the hidden character lengths where `radmaxsph`/`pseudofacsph` are expected). The reference was therefore produced with an explicit `INTERFACE` block for `ctype` added to the program (no other change; absent arguments mean `radmaxsphere` = `0.5*sigcor`, `pseudofacmol` = `1`), built by `rust/tools/build_patched_readxyz_readpdb.sh`; the comparison is against this patched build, not the original binary.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.